University

Course Registration and Academic Records System

Functional requirements:

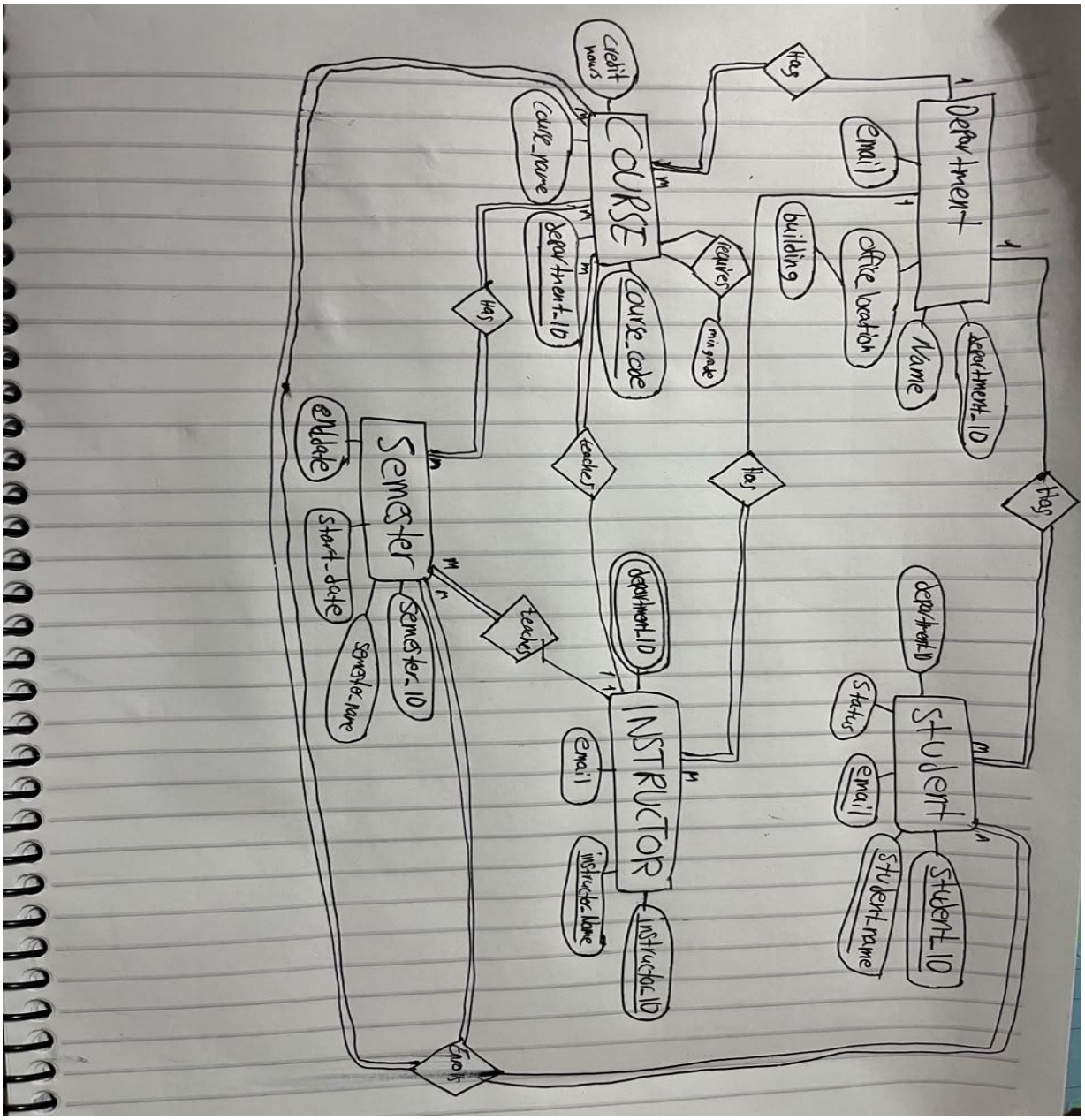
| ID | MoSCoW | requirement | Brief justification |
|---|---|---|---|
| FUN_1 | must | The database shall create, update, retrieve, and deactivate student profiles containing identity, contact, admission, program, and status information. | Student records are the foundation of registration and academic services. |
| FUN_2 | must | The database shall maintain university department records. | Departments organize programs, courses, and instructors. |
| FUN_3 | must | The database shall maintain academic programs, degree levels, and required credit totals. | Students must be evaluated against the correct degree requirements. |
| FUN_4 | Could | The database shall record attendance for each enrolled student by class date. | Attendance may support course policies and early intervention. |
| FUN_5 | must | The database shall maintain instructor profiles and their department affiliations. | Sections and grades must be connected to responsible instructors. |
| FUN_6 | must | The database shall maintain a course catalog containing course codes, titles, descriptions, departments, and credit hours. | Registration and transcripts require consistent course definitions. |
| FUN_7 | must | The database shall associate programs with their required, elective, and optional courses. | This supports curriculum planning and degree-progress checking. |
| FUN_8 | must | The database shall store course prerequisite rules and the minimum acceptable grade for each prerequisite. | Students should enter courses only when academically prepared. |
| FUN_9 | must | The database shall maintain academic terms with start, end, registration, add, and drop dates. | Transactions must follow the correct semester and deadlines. |
| FUN_10 | must | The database shall create course sections for specific courses and academic terms. | Students enroll in a scheduled section rather than the general course. |

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| FUN_11 | must | The database shall assign a primary instructor to each course section. | Teaching and grade entry responsibility must be identifiable. |
| FUN_12 | must | The database shall store each section's meeting days, times, rooms, and capacity. | Scheduling and room allocation depend on this information |
| FUN_13 | should | The database shall support searching sections by term, department, course, instructor, and meeting time. | Students and staff need to find suitable offerings efficiently. |
| FUN_14 | should | The database shall maintain an ordered waitlist and allow promotion when a seat becomes available. | A waitlist distributes limited seats fairly. |
| FUN_15 | should | The database shall assign academic advisors and retrieve each advisor's students. | Advisors need access to their assigned students for academic planning. |
| FUN_16 | should | The database shall compare completed courses with program requirements and identify remaining requirements. | Students need accurate information about progress toward graduation. |
| FUN_17 | should | The database shall produce section rosters, enrollment totals, grade distributions, and graduation eligibility reports. | University staff require consolidated information for decisions and planning. |
| FUN_18 | should | The database shall record invoice payments and calculate remaining balances. | Students and finance staff need accurate balance information. |
| FUN_19 | should | The database shall generate student invoices or tuition charges for an academic term. | Financial obligations must be associated with the correct student and term. |
| FUN_20 | must | The database shall calculate term and cumulative GPA and generate student transcripts. | Progression and graduation decisions depend on academic history. |

Non-functional requirements:

| ID | MoSCoW | requirement | Brief justification |
|---|---|---|---|
| NON_FUN_1 | must | The database shall enforce role based, least privilege access for students, instructors, registrars, finance staff, and administrators. | Academic, personal, and financial information must only be accessed by authorized users. |
| NON_FUN_2 | must | Sensitive information shall be encrypted in transit and at rest using accepted security standards. | Encryption reduces the impact of interception or unauthorized storage access. |
| NON_FUN_3 | must | Registration, grade, invoice, and payment operations shall use ACID transactions together with primary-key, foreign-key, unique, and validation constraints. | Partial or inconsistent transactions could corrupt important records. |
| NON_FUN_4 | must | At least 95% of ordinary searches shall complete within two seconds and registration transactions within three seconds under an assumed load of 500 concurrent users. | Registration periods produce heavy demand and users need timely responses. |
| NON_FUN_5 | should | The design shall support at least 50,000 active students and five million historical enrollment records without redesigning the schema. | University information grows continuously across academic years. |

ERD:


Department
department_ID
Name
office location
building
email
Has
STUDENT
student_ID
student_name
email
status
department ID
INSTRUCTOR
instructor_ID
instructor_name
email
department_ID
COURSE
course_code
course_name
credit hours
department_ID
prerequisite
requires
Has
teaches
Has
Semester
Semester_ID
Semester_name
start_date
endDate
teaches
Enrolls